FloodGuard AI

Case Study :

Disaster Resillience

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TABLE OF CONTENTS

INTRODUCTION....3
LITERATURE REVIEW....4
METHODOLOGY....5
1. System Architecture....5
2. System Features And Possible Solutions....5
AI ACKNOWLEDGEMENT....8
1. AI implementation for Flood Prediction and Alert System....8
2. AI-Powered Disaster Response Guidance System....8
3. AI-Powered Location-Based Disaster News Updates....9
PROTOTYPE OVERVIEW....10
RESULTS....11
1. Early Warning and Alert Logic (Flood Prediction Mechanism)....11
2. Evacuation Guidance....11
3. Aid Distribution Management....11
4. SOS Emergency Communication....11
5. Information Integrity....12
6. User Interaction and Persona-Based Functionality....12
CONCLUSION....13

INTRODUCTION

The Southeast Asia (ASEAN) region currently faces a period of extreme weather conditions. This is a condition where a warmer atmosphere holds more moisture leading to a more frequent and heavy rain. In the late 2025, both South and Southeast Asia experienced catastrophic flooding which claimed the lives of hundreds of people. In Indonesia alone, over 860 people died and 570,000 were displaced after the Tropical Cyclone Senyar hit Sumatra. At the same period, Thailand reported 178 deaths at least as rainfall reached levels that can only be seen once every 300 years. These statistics are a major problem as rural communities and school students are highly vulnerable because of the lack of precise data and location which are crucial for fast rescue operations.

Most of the current systems suffer from a “Lead-Time Gap”, where manual data collection and slow computer models did not manage to warn people about the flash floods fast enough. There is also a lack of information that is specific for different groups of people, which can create confusion during a crisis. FloodGuard is an AI-powered mobile app created to fix these specific issues. It shifts disaster management from just reacting to a more proactive culture of safety, which is one of the main objectives of the ASEAN Capacity Building Roadmap 2025-2030.

By using the live data from the river sensors and weather updates, FloodGuard AI model analyzes risks in real-time. The application is split for two main users, the victims and rescuers. Victims get personalized risk alerts and an SOS button to send their location to get help. Rescuers get a dashboard with live status on the victims who need help and where supplies are located. This ensures that official, verified information will reach the right people immediately. This will contribute to helping everyone during a disaster.

LITERATURE REVIEW

The development of a smart flood management system like FloodGuard AI makes the understanding of the limitations in current disaster response frameworks easier. First of all, we need to be clear on how the new technology can be helpful and why the current methods are failing. This review will examine the catastrophic socio-economic impacts of flood within the ASEAN region, the main problems with the current rescue efforts, and how Artificial Intelligence (AI) and mobile apps can integrate to create a safer future.

The Socio-Economic Impacts of Floods in the ASEAN region

Flooding remains the most common and damaging natural hazard across Southeast Asia, particularly in Malaysia. Data from the Department of Statistics Malaysia (DOSM) shows that flood losses in 2023 alone escalated to RM800 million, primarily destroying the residential infrastructure and essential public services. Aside from the money, the human cost is very high. Research indicates that people living in rural areas and school children face higher risk because they often cannot evacuate quickly. In 2025, over 1500 fatalities were recorded due to the extreme rain in Indonesia, Thailand and Philippines. Millions of people are forced to leave their homes. These high numbers show clearly that our traditional way of managing flood disaster is no longer relevant today.

Problems in Current Rescue Efforts

Despite the efforts of agencies such as NADMA and JPS, there are still big “last-mile” challenges that stop early warnings from being translated into effective action. One major issue that we noticed is the Lead-Time Gap. Traditional hydrological models are too slow to predict the “flash floods” that can occur within a few minutes. This technical limitation is compounded by poor communications, where many of the victims get alerts too late and are unable to process what actions to take when the alerts are received. Additionally, the internet can also cause “Information Overload” where tons of fake news take over the social media, making it harder for the people in need to trust official news. Research also shows that food and supplies remained unequal due to a lack of real-time data regarding the resource allocation and victim needs.

How AI and Mobile Apps Can Help

To bridge these gaps, recent academic discourse emphasizes the role of AI and mobile apps in modernizing disaster resilience. By analyzing the rain patterns and river levels, AI can predict “hyper-local” flood hours long before they actually happen. This aligns with the goals of ASEAN Capacity Building Roadmap 2025-2030. Beyond that, mobile apps also allow for Role-Based UI/UX which can show different things to different people. In this case, victims receive localized safety guides, while rescuers access high-level command dashboards to get the bigger picture. By using AI to track supplies and discover dangerous zones, FloodGuard AI can help society to act early rather than waiting for the water to rise.

METHODOLOGY

1. System Architecture

The FloodGuard application is built using a layered system architecture that integrates real-time environmental monitoring with user-centered disaster response services. The purpose of this architecture is to ensure that all flood-related information can be collected, analyzed and delivered efficiently to different types of users during emergencies.

Starting with a data acquisition layer, the system collects data from multiple sources. To ensure regional relevance, the FloodGuard system incorporates disaster-related datasets from ASEAN monitoring platforms. Historical flood records and regional disaster statistics are obtained from the ASEAN Coordinating Centre for Humanitarian Assistance on Disaster Management through the ASEAN Disaster Information Network (ADINET). This database provides verified information on disaster occurrences across ASEAN member states, including event location, disaster type, and impact reports. In addition, real-time environmental data is integrated from meteorological services and weather APIs such as OpenWeartherMap, river level monitoring system from Department of Irrigation and Drainage Malaysia (JPS) and official announcement from verified government channels across all asean countries. These data sources provide continuous updates regarding the rainfall levels, river condition and emergency advisories.

The collected data is then processed through the AI processing layer, which contains the system's Risk Analysis Model. This model evaluates the environmental indicators such as rainfall intensity, water level fluctuations and historical flood patterns to estimate the likelihood of flooding by generating a flood risk index that categorizes the situation into Low, Medium and High risk levels. In addition, the system calculates logistical information such as evacuation center proximity and resource availability.

Next, the application layer delivers the processed information through a persona-based interface. When users enter the application, they can select either as flood victim or rescuer. Based on this role, the system dynamically presents relevant features and data such as the victims can receive evacuation guidance and shelter availability while the rescue can access operational information such as rescue requests and aid distribution management.

2. System Features And Possible Solutions

2.1 Early Warning and Alert Logic (Flood Prediction Mechanism)

One of the primary objectives of FloodGuard is to provide early warnings before flooding occurs, allowing residents to prepare and evacuate safely.

The system employs a predictive analysis mechanism in which the AI model analyzes rainfall intensity and river water levels. Using these inputs, the model generates a flood risk classification

that estimates the probability of flooding in a specific area. This predictive capability helps users respond before conditions become critical.

To maintain accuracy, the system also performs real-time environmental monitoring. Weather-related metrics such as rainfall volume, river water levels, and humidity are refreshed automatically every five minutes. This ensures that users receive updated information as conditions evolve.

When the calculated risk index reaches a high-risk threshold (for example, a score above 70 on a scale of 0–100), the system activates an emergency alert mechanism. The alert appears on the user's dashboard as a pulsing warning banner accompanied by an audible notification. This multi-sensory alert system is designed to quickly capture user attention and prompt immediate action.

2.2 Evacuation and aid distribution logic

a. Shelter finder logic

During floods, victims must quickly locate safe evacuation centers that are accessible and not overcrowded. FloodGuard simplifies this process through an intelligent Shelter Finder system.

First, the application automatically detects the user's current location using GPS-based positioning. The system then retrieves a list of registered evacuation centers and sorts them based on real-time travel distance. This ensures that victims are directed to the nearest and most accessible shelters.

Additionally, each evacuation center continuously updates its live occupancy data whenever new evacuees arrive. This information is reflected on the system's map interface through a Live Capacity Indicator, allowing users to see which shelters still have available space. By providing this real-time information, the system prevents victims from traveling to overcrowded centers.

b. Aid distribution logic

Efficient distribution of humanitarian aid is essential during flood emergencies. FloodGuard incorporates an Aid Distribution Monitoring System that enhances transparency and coordination.

Evacuation centers and aid providers can register their available resources, including essential supplies such as food, drinking water, and medical items. These inventories are tracked through a centralized dashboard that displays current supply levels.

To ensure fair distribution, each aid request submitted by victims is recorded with a timestamp and geographic location. This allows rescuers to prioritize requests

systematically based on urgency and proximity, preventing unequal or inefficient allocation of resources.

Furthermore, the system includes a low-stock monitoring function. When the quantity of any supply category drops below 20% of its capacity, the system automatically generates an alert to notify coordinators that additional resources are required.

c. SOS Emergency Broadcast

The Emergency Screen is designed to help flood victims quickly contact emergency responders and request rescue assistance during critical situations. In flood disasters, victims may experience panic, limited mobility, or poor communication access. Therefore, this feature simplifies the process of reaching rescuers with just one action.

The main feature of this screen is the "Tap to Send SOS" button. When the user taps this button, the system automatically sends an emergency distress signal containing the user's current GPS location to nearby emergency services or rescue authorities.

This allows rescue teams to quickly identify the victim's location and respond faster, especially in situations where the victim cannot clearly explain their location.

Each contact includes a one-tap call button, allowing users to immediately call the relevant emergency service without needing to search for phone numbers.

This feature supports the evacuation process by ensuring that victims can quickly communicate with rescuers, request help, and receive guidance during flood emergencies. By combining an SOS alert system with direct emergency contacts, the application reduces response time and improves the chances of safe evacuation.

2.3 Information integrity and misinformation prevention

During disasters, misinformation can spread rapidly and create confusion among affected communities. To address this challenge, FloodGuard includes a Trusted Information Hub designed to ensure the reliability of shared information.

All official content within this section is sourced exclusively from verified government agencies, including National Disaster Management Agency (NADMA), Department of Irrigation and Drainage Malaysia (JPS), and Malaysian Meteorological Department (MetMalaysia).

Before any article or advisory appears on the platform, it must pass a source validation process to confirm that the information originates from an authorized agency. Once verified, the content is marked with a "Verified" badge to indicate its authenticity.

Importantly, this section does not allow user-generated content. By restricting contributions to official sources only, the system eliminates the risk of rumors, unverified claims, or fake news spreading through the application.

AI ACKNOWLEDGEMENT

1. **AI implementation for Flood Prediction and Alert System**

   The implementation of Artificial Intelligence (AI) in the flood alert system begins with collecting real-time environmental data using a weather Application Programming Interface (API). The system sends periodic requests to the weather API to obtain updated meteorological information such as rainfall intensity, humidity, temperature, wind speed, and atmospheric pressure. This data is usually returned in JSON format and becomes the primary input for the prediction model. In addition to real-time weather data, the system may also incorporate historical flood records and environmental information such as river water levels and soil moisture to improve prediction accuracy.

   Before the data is used for analysis, it undergoes a preprocessing stage where incomplete or inconsistent values are cleaned, missing data is handled, and all variables are normalized into consistent units. Important features such as rainfall amount, rainfall duration, river level changes, and soil saturation are then extracted as key indicators that influence flood occurrence. After preprocessing, the prepared dataset is fed into a machine learning model that has been trained using historical flood data. Algorithms such as decision trees, random forest, or neural networks can be used to identify patterns between weather conditions and previous flood events. Through this training process, the AI model learns the relationships between environmental variables and the likelihood of flooding. When real-time weather data is received from the API, the trained AI model analyzes the input and calculates the probability of a flood occurring in a particular area. The system then classifies this probability into risk categories such as low, medium, or high flood risk based on predefined thresholds. If the prediction indicates a medium or high flood risk, the system automatically activates the alert mechanism. Early warning notifications are then sent to users through the mobile application, SMS messages, or dashboard alerts to inform them about the potential flood risk and provide time for preparation or evacuation.

   Additionally, all newly collected weather data and prediction outcomes are stored in the system database, allowing the AI model to be retrained periodically with updated data. This continuous learning process helps improve prediction accuracy over time and enables the system to adapt to changing climate patterns and environmental conditions. Through the integration of real-time weather APIs, machine learning algorithms, and automated notification systems, the proposed solution provides a proactive flood early warning mechanism that enhances community preparedness and reduces the potential impact of flooding disasters.

2. **AI-Powered Disaster Response Guidance System**

   The Safety Guide feature in the mobile application uses Artificial Intelligence (AI) to provide real-time recommendations and a dynamic to-do list for users during a natural disaster such as a flood. When the system detects a potential flood risk through the AI prediction model and real-time weather data obtained from external weather APIs, the application automatically activates the emergency safety interface. The AI analyzes the current situation by considering several factors such as predicted flood severity, water level risk, user location, and nearby safe

zones. Based on this analysis, the system generates personalized safety suggestions that guide users on what actions to take before, during, or after the disaster. For example, during a flood emergency the AI may recommend immediate actions such as evacuating the area, avoiding electrical hazards, switching off the main power supply, and unplugging electrical appliances to reduce the risk of accidents.

The system also automatically updates the user's readiness progress by tracking completed safety steps, helping users quickly identify which important actions still need to be taken. By presenting these instructions as a structured checklist, the application reduces panic and helps users respond more effectively in emergency situations. This intelligent safety assistance transforms raw disaster data into practical and actionable guidance, ensuring that users receive timely instructions that improve preparedness, safety, and decision-making during natural disasters.

3. AI-Powered Location-Based Disaster News Updates

The Latest News feature in the application uses Artificial Intelligence (AI) to provide users with real-time disaster-related news and updates based on their current country or geographic location. The system collects information from trusted online news APIs and official disaster management sources that continuously publish updates about natural disasters, weather warnings, evacuation notices, and emergency announcements.

AI is used to filter, analyze, and prioritize the most relevant news articles by identifying keywords related to disasters such as floods, storms, landslides, or heavy rainfall. In addition, the system uses the user's location data to ensure that the information shown is relevant to their country or nearby regions. For example, if a flood warning is issued in the user's area, the AI system will prioritize displaying news reports and official updates related to that event. Natural Language Processing (NLP) techniques can also be used to summarize long news articles into short and easy-to-understand updates so that users can quickly grasp important information during an emergency situation. By automatically collecting and filtering the latest disaster-related news, this feature helps users stay informed about ongoing situations, government advisories, evacuation instructions, and safety measures. As a result, the AI-powered news update system improves situational awareness and ensures that users receive timely and location-relevant information during natural disasters.

PROTOTYPE OVERVIEW

FloodGuard AI is a high-fidelity interactive model developed using Figma. The goal of this prototype is to demonstrate the user journey and the visual logic part of the app's features. By using Figma's advanced prototyping tools, we have simulated a real-world experience that shows how a user would interact with the system during a flood crisis. This will allow us to test the usability and navigation of the app before we begin any actual coding.

The design of this prototype revolves around two main flows based on the user's role alongside interactive logic and information integrity.

1. **The Civilian Flow (Mangsa/Resident)**

   Designed for personal safety, this path centers on an AI-calculated flood risk index displayed on the main dashboard. The interactive flow allows victims to navigate from real-time weather alerts to an Evacuation Centre Finder that uses GPS data to sort the nearest shelters by driving distance. It also includes a Safety Guide with interactive checklists and an Emergency Contacts screen featuring a prominent SOS broadcast button.

2. **The Responder Flow (Penyelamat/Rescuer)**

   This flow transforms the app into a Command Dashboard for response teams. The prototype demonstrates advanced coordination tools, such as Neighbourhood Monitoring for community-powered safety reports and an Aid Distribution Tracker. The latter uses AI to flag when supply levels for food, water, or medicine drop below 20%

3. **Interactive Logic and Information Integrity**

   A critical component of the Figma prototype is the Trusted Info Hub, which simulates a curated feed of verified news from official agencies like NADMA, JPS, and PDRM. To combat misinformation, the design features color-coded source badges and a verified indicator, strictly prohibiting user-submitted content in this section.The visual design emphasizes speed and reliability, with critical information engineered to be visible in under three seconds. By using Smart Animate and component states, the prototype illustrates how the layered architecture which connects live Weather APIs and JPS river sensors, delivers a personalized, proactive safety culture to the user's fingertips.

RESULTS

1. **Early Warning and Alert Logic (Flood Prediction Mechanism)**

   The FloodGuard prototype successfully achieved its objective of providing early flood warnings by generating a flood risk index categorized as Low, Medium, or High. The AI-based predictive model analyzed simulated rainfall intensity and river water levels to estimate the probability of flooding in targeted areas. Alerts were activated when the calculated risk index exceeded the designated threshold, appearing on the user dashboard with both visual and audible notifications. Testing demonstrated that the system could deliver alerts within one second of threshold crossing, while environmental data was refreshed every five minutes, ensuring that users consistently received accurate, real-time updates. These results indicate that the system is capable of providing timely early warnings, thereby allowing users to take precautionary measures before flood conditions become critical.

2. Evacuation Guidance

The Shelter Finder feature of FloodGuard effectively facilitated safe evacuation by identifying the nearest accessible evacuation centers based on the user's GPS location. During simulation scenarios, the system successfully sorted shelters according to travel distance and presented live occupancy data to prevent overcrowding. Test results confirmed that users were able to select appropriate shelters quickly and efficiently, demonstrating the system's ability to guide victims to safe locations. This confirms that the prototype effectively addresses the challenge of rapid evacuation planning during flood events.

3. Aid Distribution Management

The aid distribution mechanism within FloodGuard demonstrated the ability to track and manage essential resources such as food, water, and medical supplies. Each aid request submitted by users was recorded with a timestamp and geographic location, enabling rescuers to prioritize distribution based on urgency and proximity. The low-stock monitoring function accurately triggered alerts when resource levels fell below 20% of capacity, ensuring timely replenishment. These findings indicate that the prototype can support organized and equitable allocation of humanitarian aid during emergencies, enhancing operational efficiency and fairness.

4. SOS Emergency Communication

The Emergency Screen successfully facilitated direct communication between flood victims and rescue authorities. Users could send an SOS alert with their precise GPS location through a single tap, while one-tap call buttons enabled immediate contact with emergency services. Simulation tests confirmed that the distress signals were delivered accurately and in a timely manner. These results suggest that the system can significantly reduce response times and improve the safety of users in critical situations where immediate assistance is required.

5. Information Integrity

FloodGuard's Trusted Information Hub maintained the reliability and authenticity of information presented to users. All content was sourced exclusively from verified government agencies, including NADMA, JPS, and MetMalaysia, and underwent a validation process before being displayed. During testing, all information appeared with a "Verified" badge, and no unverified or user-generated content was present. This confirms that the system effectively prevents the spread of misinformation during flood events, thereby enhancing user trust and ensuring access to credible updates.

6. User Interaction and Persona-Based Functionality

The prototype demonstrated successful implementation of persona-based interfaces. Flood victims were provided with evacuation guidance, shelter availability, and aid request functionalities, whereas rescuers were equipped with tools to monitor operations, coordinate aid, and manage rescue requests. The results indicate that the system can dynamically adapt its content and features according to user roles, ensuring that relevant information is delivered efficiently to support decision-making during emergencies.

CONCLUSION

The development of FloodGuard AI represents a significant step forward in modernizing disaster management for the ASEAN region. By addressing the critical "Lead-Time Gap" and the "last-mile" communication challenges, this project changes the focus from traditional reactive measures to a proactive culture of safety. The integration of real-time environmental data with advanced Artificial Intelligence allows for hyper-local predictions that give residents the significant time they need to evacuate before conditions become life-threatening. Through the use of role-based interfaces, the application ensures that both victims and rescuers receive only the most relevant, actionable information, hence reducing confusion and information overload especially during high-stress situations.

The prototype results demonstrate that a centralized, AI-powered platform can effectively coordinate complex logistics such as shelter finding and aid distribution. By making the monitoring of supply levels automatic and providing live occupancy data for evacuation centers, FloodGuard AI helps prevent the unequal distribution of resources and overcrowding at shelters. Furthermore, the inclusion of the Trusted Info Hub serves as an absolutely critical tool in maintaining information integrity, ensuring that communities are guided by verified government data rather than dangerous rumors or misinformation.

Ultimately, FloodGuard AI aligns with the strategic goals of the ASEAN Capacity Building Roadmap 2025–2030 by leveraging mobile technology to build regional disaster resilience. The project proves that when technical innovation is paired with a deep understanding of human needs, it is possible to create a system that can predict disasters and also provides a clear, safe path through them at the same time. As climate conditions continue to become more extreme, tools like FloodGuard will be essential in protecting vulnerable populations, saving lives, and reducing the huge socio-economic impacts of flooding across the region.